

Week 4, September 23, 2026

Main topics

- Tikhonov regularization
- Inverse Problems and Imaging beyond Tikhonov

Material

From last week: Kirsch 2.2. Pay attention to the equivalence of (2.15), (2.16) and (2.19).
No new material this week

13:00-13:30 Tikhonov regularization

13:30-15:30 Exercises, see below.

15:45-17:00 Research lecture: Yiqiu Dong on Inverse Problems and Imaging beyond Tikhonov

Exercises

You are highly encouraged to discuss and solve the exercises in small groups. If you are not very comfortable with infinite dimensional analysis (functional analysis) then you can think of a matrix $A \in \mathbb{R}^{n \times m}$ and vector $y \in \mathbb{R}^n$ whenever you encounter K between Hilbert spaces.

1. For TSVD and Tikhonov regularization the filter factors are given by

$$q(\mu) = \begin{cases} 1 & \mu > \alpha^2 \\ 0 & \mu \leq \alpha^2 \end{cases}, \quad q(\mu) = \frac{\mu^2}{\mu^2 + \alpha}.$$

Plot the filter factors (in loglog) for both TSVD and Tikhonov regularization for different truncations and regularization parameters α . Discuss pros and cons for using one or the other.

2. By calculating the 2-norm, show that for a matrix $A \in \mathbb{R}^{m \times n}$, vector $y \in \mathbb{R}^m$ and $\alpha > 0$

$$\|Ax - y\|_2^2 + \alpha\|x\|_2^2 = \left\| \begin{bmatrix} A \\ \sqrt{\alpha}I \end{bmatrix} x - \begin{bmatrix} y \\ 0 \end{bmatrix} \right\|_2^2,$$

with $I \in \mathbb{R}^{n \times n}$ denoting the identity matrix and $0 \in \mathbb{R}^n$. Conclude that the Tikhonov solution can be obtained by minimizing the functional

$$J_\alpha(x) = \left\| \begin{bmatrix} A \\ \sqrt{\alpha}I \end{bmatrix} x - \begin{bmatrix} y \\ 0 \end{bmatrix} \right\|_2^2.$$

3. Show that for a compact operator K between Hilbert spaces X and Y and $\alpha > 0$, $(K^*K + \alpha I)$ is invertible:

- Show first that the operator is injective.
 - If you are not comfortable with functional analysis, consider $K = A \in \mathbb{R}^{m \times n}$ and draw the conclusion from linear algebra.
 - If you are familiar with functional analysis, invoke the Fredholm alternative.
4. Consider the Abel model problem and generate for $n = 100$ as in previous weeks the matrix A , the sampled function x , the righthand side $y = Ax$ and noisy $y^\delta = y + d$, with relative noise level $\|d\|/\|y\| = \delta$.
- Fix $\delta = 0.01$ and compute (using the least squares formulation from exercise 1.) the Tikhonov regularized solution x_α for $\alpha = 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 10^0$. Plot the reconstructions and comment on the result.
 - Repeat the above for $\delta = 0.05$. What is the optimal regularization parameter α (what does optimal actually mean)?
5. The compact operator $K : X \rightarrow Y$ has singular system $\{x_j\}, \{y_j\}, \mu_j$. Use the fact that

$$\begin{aligned}(K^*K + \alpha I)x_j &= (\mu_j^2 + \alpha)x_j \\ K^*y_j &= \mu_j x_j\end{aligned}$$

to derive the Tikhonov reconstruction formula

$$x_\alpha = \sum_{j=1} \frac{\mu_j}{\mu_j^2 + \alpha} (y_\delta, y_j) x_j.$$

6. For the backwards heat problem ($T = 0.1$), create for $M = 100$ the matrix. Illustrate in the same plot (loglog) the svd factors ($\mu_j/q(\mu_j)$) for Tikhonov ($\alpha = 10^{-4}$), TSVD ($k = 5$). Along the first axis, put $\log \mu_j$ and out the second axis, the filter factors.
7. Show that the Tikhonov functional

$$J_\alpha(x) = \|Kx - y^\delta\|^2 + \alpha \|x\|^2$$

is strictly convex. Use this fact to prove uniqueness of a minimizer of the functional. If you are not comfortable with functional analysis, then take $K = A \in \mathbb{R}^{n \times m}$ and $y^\delta \in \mathbb{R}^n$.